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Tracks for Enhancing Rainfall-Induced Landslide Prediction

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1. Introduction and Objectives

Rainfall-induced landslides represent a growing climate-related hazard, motivating the development of predictive models that can effectively leverage increasing volumes of high-resolution environmental data. Traditionally, landslide prediction has followed either a process-based, geomorphological perspective—focused on local physical mechanisms—or a statistical perspective based on regional inventories and empirical relationships. Over time, the latter has evolved from simple correlations toward increasingly sophisticated data-driven approaches, including machine learning methods [2].

Most operational early warning systems for rainfall-induced landslides rely on rainfall intensity–duration thresholds, which identify conditions historically associated with slope failure [3]. While effective and interpretable, these static thresholds struggle to capture the complex temporal structure of rainfall and antecedent wetness, and often require regional recalibration to account for climatic and geomorphological variability, as illustrated by applications in Switzerland [4].

Recent advances in climate informatics and machine learning offer new opportunities to improve landslide forecasting by moving beyond threshold-based approaches. Neural networks and sequence models have demonstrated strong performance in climate and hydrological applications [1], enabling probabilistic predictions that explicitly exploit temporal dependencies. In this context, we compare event-based and time-series-based machine learning models for rainfall-induced landslide prediction, evaluating both predictive skill and model explainability, with particular attention to the role of time-lagged rainfall information.

2. Methods

Landslide events were obtained from the damage-based dataset by the Swiss Federal Institute for Forest, Snow and Landscape (WSL). Hourly precipitation data were obtained from the CombiPrecip product of MeteoSwiss, which combines radar and rain-gauge measurements into a spatially continuous 1 × 1 km gridded dataset over Switzerland. The analysis covers the period 2002–2024. Precipitation time series were extracted for all grid cells affected by recorded landslides. Rainfall events were reconstructed using a Python adaptation of the CTRL-T algorithm developed at IRPI, which identifies independent precipitation episodes based on minimum inter-event dry periods. Pseudo-negative samples were defined as rainfall events occurring at landslide-prone grid cells that did not result in slope failure, ensuring realistic hydrometeorological variability while minimizing spatial bias.

For event-based prediction, each rainfall event was divided into antecedent and triggering periods following previous studies. The triggering period was defined as the 24 hours surrounding the landslide occurrence (or the end of the event for pseudo-negatives), while antecedent conditions were summarized

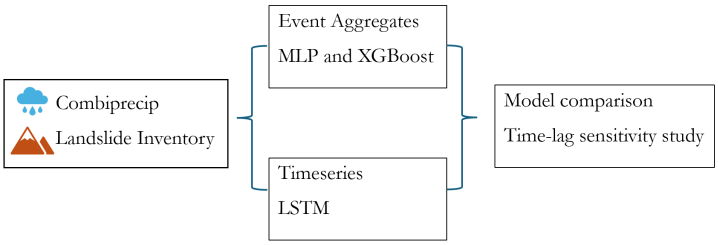

Figure 1. *Schematic summary.*

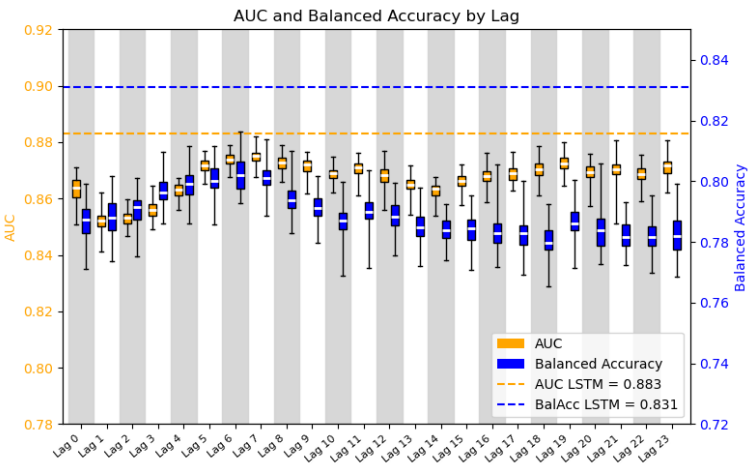

Figure 2. *Model performance (AUC and Balanced Accuracy) of MLP for timelags 1 to 24 and median LSTM result for 400-hour series length. Spread represents the different ensembles. Higher is Better..*

through cumulative rainfall and duration. For time series-based analysis, the original time series was conserved.

Two families of classification models were employed. Event-based models included XGBoost and a multi-layer perceptron (MLP) trained on aggregated rainfall features. To address class imbalance, a bagging strategy was applied by repeatedly resampling pseudo-negative events to match the number of positives, yielding probabilistic ensemble predictions. In parallel, a Long Short-Term Memory (LSTM) neural network was trained on full hourly precipitation sequences preceding each event to explicitly capture temporal dependencies. Variants of this architecture were then trained and compared with different lead times between 24 an 800 hours, with and without antecedent conditions, modeled as cumulated sums of precipitation.

3. Preliminary Results

Figure 2 shows preliminary results of MLP vs LSTM performance. MLP is expected to perform worse since it relies on scalar aggregates whereas the LSTM disposes of richer data. Other results might include visualizations of prediction, and a scan over LSTM timeseries length. As a part of discussion, we also plan to assess the liability of observations made with CombiPrecip in contrast with Swiss precipitation forecasting systems.

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